

Qi Jia (Jerry) Sun

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EDUCATION

Ph.D.

University of Illinois Urbana-Champaign

Aug 2026–Present

Ph.D. Student in Natural Resources and Environmental Science

Agroecosystem Sustainability Center *Advisor*: Professor Kaiyu Guan

Undergraduate

University of California, Berkeley

August 2020–Dec 2024

Bachelor of Arts in Computer Science and Environmental Science

Selected coursework: Information Devices, Neural Networks, Optimization, Discrete Math, Time Series, Efficient Algorithms, Intractable Programs, Machine Learning, Artificial Intelligence, Forecasting

Activities: Teaching Assistant, CS 188 Artificial Intelligence; Biometeorology Lab; Berkeley Consulting

WORK EXPERIENCE

Google LLC

May 2025 - Present

Software Engineer

- Built a versioned rule configuration system in Protoconf that bypassed the TPC proxy, cutting latency by 96% while meeting data-sovereignty requirements; launched for 4 teams 50% ahead of schedule.
- Led the AutoFlex CUD launch across 10 servers in TPC Paris and Germany for approximately 5,000 customers; configured GMon and custom alerts to track rollout health and surface deployment issues.
- Designed the TPC credits pipeline to enable credits with Annealing, then enabled its servers for rollout.
- Automated server turnup, rollout monitoring, refactored Sisyphus workflows, and expanded the Zamm deployment playbook to reduce operational turmoil for engineers in Google Cloud Business Platform

Susquehanna International Group

June 2023 - August 2023

Quantitative Trading Intern

- Built an algorithm to find profitable FX currency pairs using moving averages and value decay durations.
- Built a web app to visualize decay in variance and beta among FX pairs following currency events.
- Designed and created a CI/CD system for 8 FX traders that provided real-time financial data updates.

Smartsheet

May 2022 - Aug 2022

Software Engineering Intern

- Reduced overloading by 35% by identifying 7 key indicators over stages of the blueprint creation process
- Devised a mathematical formula to calculate blueprint health to decrease blueprint latency by 24%
- Created user-friendly UI to integrate tool into existing dashboard for the use of over 2 million users

RESEARCH EXPERIENCE

Rice Rotation Modeling with ECOSYS and KGML (Current Ph.D. Project)

- Completed a source-traced rice rotation dataset for 353 counties, linking 5.7 million field-year records from 2008–2025 with crop histories, planting and harvest dates, and modeled fertilizer inputs and tillage.
- Validated ECOSYS simulations against satellite GPP estimates, rice-field ET measurements, and yield.
- Designed a rice-specific KGML surrogate trained on ECOSYS simulations to quantify how crop rotations, nitrogen management, and flooding affect rice yield and soil carbon across different soils and climates.

Perspective: Shifting the focus from invasive species to novel communities with remote sensing

Global Ecology and Biogeography

- Built workflows for ground and NEON airborne spectra from 4 grassland sites, using continuum removal and shared PCA to calculate spectral convex-hull volumes, comparing spectral and community variation.
- Co-authored a perspective on identifying invasive plant communities with hyperspectral data at Konza.

SELECTED ACADEMIC ACHIEVEMENTS

- **International Geography Olympiad** - Ranked 75/200 (2019), Canadian T4 Winner (2019, 2020)
- **IISEF Team Canada Top 30** - Fake News Detection ([2019](#)), El Nino Forecasting ([2020](#))
- **USA Computing Olympiad (USACO) Gold Division** - Top 800 participants (2019, 2020)
- **American Invitational Mathematics Exam** - Top 2.5% participants (2018, 2019, 2020)